

Data Communication & Networks

Complete Exam Cheat Sheet
the lecturer · Rivers State University

5 Components

Fourier Series & Transform

OSI 7 Layers

Network Topologies

Switching & Routing

Lecturer
the lecturer

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Data Communication

*The **sharing/transfer of information** locally or remotely between two or more devices using computing and communication technologies.*

3 Fundamental Principles

- 1 **Delivery** - Data must reach the **correct destination/recipient**
- 2 **Accuracy** - Data delivered **without alteration**; incomplete data = unusable
- 3 **Timeliness** - Data must be delivered **on time**; late delivery = ineffective

5 Components of a Data Communication System

- 1 Message** - The actual data to be transmitted
Text, numbers, pictures, audio, video
- 2 Sender** - Originating device that sends the data
Computer, phone, camera
- 3 Receiver** - Destination device that receives the message
TV, phone, workstation
- 4 Transmission Medium** - Physical path from sender to receiver
Twisted pair, coaxial, fibre-optic, radio waves
- 5 Protocol** - Set of rules governing communication; enables synchronization
HTTP, HTTPS, FTP, SSH

*Without a protocol: two devices may be connected but **not communicating** · like a French speaker talking to a Japanese-only speaker.*

Data Representation

All data = bit patterns (0s and 1s)

TYPE	HOW REPRESENTED	KEY DETAIL
Text	Bit patterns (codes)	Unicode = 32 bits/char (all languages). ASCII = Basic Latin (127 chars), subset of Unicode
Numbers	Direct binary	NOT via ASCII · converted directly to binary; includes bits for math operations
Images	Matrix of pixels	Pixel bit patterns. Colour = RGB or YCM . Black=00, D.Grey=01, L.Grey=10, White=11
Audio	Continuous signal → digital	Sampled → Quantized → Encoded to bit patterns
Video	Images + signal processing	Continuous entity OR combination of images. Must be Sampled → Quantized → Encoded

Key chain: Continuous Signal → **Sample** (measure at intervals) → **Quantize** (approximate finite values) → **Encode** (represent as bits)

Data Flow · 3 Transmission Modes

Defines the **direction** of data transmission between devices

Simplex



One direction only. Sender can only send; receiver can only receive.

Keyboard→Computer · TV Broadcast · Radio · Computer→Printer · Monitor (output only)

Half-Duplex



Both directions, but NOT simultaneously. One at a time; like a one-lane road with construction.

Walkie-talkie (say "over") · WiFi · CB Radio · USB v2 · Ethernet Hubs · Web Browser

Full Duplex



Both directions simultaneously. Channel capacity is shared between both directions.

Telephone Network · Live Chat · Router+Switch · Duplex Fibre-Optic · USB v3

Transmission Media · Guided

Cable directs signals along a specific path (also called **bound media**)

TYPE 1

Twisted Pair

Two **copper conductors** twisted together, each insulated with plastic. Number of twists per unit = quality.

STP (Shielded) · foil + copper shielding; less interference; more secure

UTP (Unshielded) · flexible; susceptible to interference; up to 1 km

Uses: Telephone, 10Base-T, 100Base-T LAN

TYPE 2

Coaxial Cable

Central copper core → insulator → outer conductor foil → plastic cover.

Layers (inside → outside):

Central Core → Insulator → Outer Conductor → Plastic Cover

Outer conductor = noise shield
AND circuit conductor

Uses: Cable TV, CCTV, Satellite install, Telephone networks

Both can be used for longer distances than twisted pair. Coaxial supports more stations on a common line.

Fibre-Optic Cable

HOW IT WORKS

Transmits signal as **light**.
Made of glass or plastic. Uses **reflection** to guide light · beam moving through core reflects off cladding due to density difference.

Layers (outside → inside):

Outer Jacket (PVC/Teflon) →
Kevlar → **Cladding** → **Core**

Uses: Internet backbone, 100Base-FX, 1000Base-X, undersea cables, medical laser surgery

6 ADVANTAGES

- ✓ High bandwidth
- ✓ Low attenuation (8–50 km vs 5 km for copper)
- ✓ Immune to EM interference
- ✓ Resistant to corrosion (glass)
- ✓ Lightweight
- ✓ Immune to tapping

3 DISADVANTAGES

- ✗ Requires expert installation
- ✗ Expensive to install
- ✗ Fragile · breaks if bent sharply

Wireless Communication

Signal travels in 3 ways: **Ground · Sky · Line-of-Sight**

Radio	3 KHz – 1 GHz · Omnidirectional · Long range (sky mode) · Multicasting
Microwave	1 – 300 GHz · Unidirectional · Line-of-sight · Parabolic dish / Horn antenna
Infrared	300 GHz – 400 THz · Cannot penetrate walls · Short range only

MEDIUM	DIRECTION	KEY EXAMPLES
Radio Waves	Omni	AM/FM radio, TV, Maritime, Cordless mics, Paging
Microwave	Uni	Terrestrial: short wireless; Satellite: TV, telephony, live news. Repeaters every 25–30 km
Infrared	Short	Remote controls, Night vision, Missile guidance, Thermal cameras, Gas detectors

Microwave key rules: Antennas *MUST* be aligned. Very high frequency cannot pass through walls. Some bands require regulatory permission.

Fourier Analysis

Decomposes **complex signals** → **simple sine & cosine waves**.

Converts data from **Time Domain** → **Frequency Domain**.

Created by **Joseph Fourier** (French mathematician & physicist).

FOURIER SERIES

For Periodic Signals

Signals that repeat from $-\infty$ to $+\infty$. Expands signal into harmonics (sin/cos with different amplitudes and frequencies).

FOURIER TRANSFORM

For Non-Periodic Signals

Real-life signals are generally non-periodic. Decomposes into oscillatory functions. No information created or lost.

4 Application Areas

ENGINEERING

Filter noise, compress signals, data analysis

PHYSICS

Acoustics, optics, quantum mechanics

NEUROSCIENCE

Find hidden rhythms in neural recordings

MATHEMATICS

Reduce complex differential equations to algebraic

Fourier Series Formula

$$f(x) = a_0/2 + \Sigma [a_n \cos(n\pi x/L) + b_n \sin(n\pi x/L)]$$

WHERE $N = 1$ TO ∞ AND L = PERIOD OF THE SIGNAL

COEFF.	FORMULA	MEANING
a_0	$(1/2L) \int f(x) dx$	Average (DC) value of the signal
a_n	$(1/L) \int f(x) \cdot \cos(n\pi x/L) dx$	Cosine coefficients
b_n	$(1/L) \int f(x) \cdot \sin(n\pi x/L) dx$	Sine coefficients

EVEN FUNCTION

$$b_n = 0$$

$$f(-x) = f(x)$$

→ **Fourier Cosine Series**

ODD FUNCTION

$$a_n = 0$$

$$f(-x) = -f(x)$$

→ **Fourier Sine Series**

Period · Frequency · Amplitude

PERIOD (T)

Distance before repeat

$$\sin(x) \rightarrow 2\pi$$

$$\sin(nx) \rightarrow 2\pi/n$$

Larger n = **shorter** period

$$\sin(2x) = \pi$$

$$\sin(3x) = 2\pi/3$$

$$\sin(5x) = 2\pi/5$$

FREQUENCY (F)

Opposite of Period

$$f = 1/T$$

$$\sin(x) \rightarrow$$

Fundamental Frequency

$$\sin(3x), \sin(5x),$$

$$\sin(7x) \dots \rightarrow$$

Harmonics

(integer multiples of fundamental)

AMPLITUDE (A)

Height of Wave

Controls the **size** of the wave.

$$y = A \cdot \sin(\theta)$$

If $A = 1 \rightarrow$ wave moves from **+1** to **-1**

Square wave approximation:

$$f(x) = (4/\pi) [\sin(x)/1 + \sin(3x)/3 + \sin(5x)/5 + \sin(7x)/7 + \dots]$$

3 Types of Fourier Series Expansion: (1) Trigonometric · (2) Complex Exponential · (3) Polar / Harmonic

Square Wave · Full Derivation

Given: $f(x) = A$ for $0 < x < \pi$ | $f(x) = -A$ for $\pi < x < 2\pi$

STEP	RESULT	WHY?
Find a_0	$a_0 = 0$	Symmetric wave · positive and negative areas cancel: $A\pi - A\pi = 0$
Find a_n	$a_n = 0$	$\sin(n\pi) = 0$ and $\sin(0) = 0$ for all integer n · everything cancels
Find b_n (even n)	$b_n = 0$	$(-1)^n = +1$ when n is even $\rightarrow 1 - 1 = 0$
Find b_n (odd n)	$b_n = 4A/n\pi$	$(-1)^n = -1$ when n is odd $\rightarrow 1 - (-1) = 2 \rightarrow b_n = A/n\pi \times 2 \times 2 = 4A/n\pi$

$$f(x) = (4A/\pi) [\sin(x) + \sin(3x)/3 + \sin(5x)/5 + \sin(7x)/7 + \dots]$$

KEY CONCLUSION: Only ODD harmonics survive. Even harmonics cancel out.

Fourier Transform

For **non-periodic** signals · no information created or lost

$$F(\omega) = \int f(t) \cdot e^{-i\omega t} dt$$

FORWARD TRANSFORM (TIME → FREQUENCY)

$$f(t) = (1/2\pi) \int F(\omega) \cdot e^{i\omega t} d\omega$$

INVERSE TRANSFORM (FREQUENCY → TIME)

Note: $e^{-i\omega t} = \cos(\omega t) - i\sin(\omega t)$ | $f(t)$ = time domain | $F(\omega)$ = frequency domain

Worked Example (Exam Answer)

Find $F(\omega)$ of $f(t) = e^{-at} \cdot u(t)$, $a > 0$ where $u(t)=1$ for $t \geq 0$, $u(t)=0$ for $t < 0$

$$F(\omega) = \int_0^\infty e^{-at} \cdot e^{-i\omega t} dt = \int_0^\infty e^{-(a+i\omega)t} dt$$

Using $\int_0^\infty e^{-bt} dt = 1/b$ (where $b = a+i\omega$, $a > 0$):

$$\mathbf{F(\omega) = 1 / (a + i\omega)}$$

Square pulse: $F(\omega) = T \cdot \text{sinc}(\omega T/2)$. Wider pulse in time → **narrower** frequency spread. Narrower pulse → **wider** spread.

Serial vs Parallel Transmission

Serial Transmission

- Bits:** One at a time
- Wires:** 1 channel
- Distance:** Long distance
- Cost:** Cheaper
- Speed:** Slower (short dist.)
- Risk:** Sync issues

USB · Ethernet · RS-232 · Satellite · WiFi · 4G/5G · Fibre backbone

Parallel Transmission

- Bits:** Multiple at once
- Wires:** Multiple (1 per bit)
- Distance:** Short distance only
- Cost:** Expensive
- Speed:** Faster
- Risk:** Signal skew

CPU data bus · Computer printer port · Internal memory transfer

Long-Distance Serial Examples

TECHNOLOGY	DETAIL
Digital Telephony	Voice sent bit by bit over T1/E1 lines
Fibre-Optic Internet	Light pulses in single stream · backbone & undersea cables
WiFi / 4G / 5G	Data packets transmitted bit by bit over radio waves

Asynchronous vs Synchronous

Asynchronous

```
| Start | D0 D1 D2 D3  
D4 D5 D6 D7 | Parity |  
Stop |
```

Unit:	Character by character
Clock:	No shared clock
Overhead:	Start + Stop bits
Error:	Parity bit
Best for:	Sporadic data

RS-232 · Keyboard · Mouse · Dial-up modem
Real-life: Short text messages

Synchronous

```
| Sync | Sync | Data |  
Data | Data | Data |  
Data |
```

Unit:	Blocks / Frames
Clock:	Shared clock / preamble
Overhead:	None (no start/stop)
Error:	Checksum / CRC
Best for:	Bulk continuous data

Ethernet · Broadband · Digital Telephony
Real-life: Phone calls, Video calls

ASync ADVANTAGES

- ✓ Simple & cost-effective
- ✓ Good for sporadic data
- ✓ Easy to implement

ASync DISADVANTAGES

- ✗ Extra bits reduce efficiency
- ✗ Slower for bulk data
- ✗ Less precise timing

Computer Networks

A collection of **interconnected devices** that communicate and share resources

LAN

Local Area Network

Office/Home

High speed, small area, inexpensive

MAN

Metropolitan Area Network

City / Large Campus
Connects nearby LANs

WAN

Wide Area Network

Global / Country
Connects LANs + MANs; Internet

LAN properties: Very high speed · Small geographical area · Less expensive technology · Share hardware/software · Send messages & emails between users

Network Topologies

Connection arrangement of nodes · defines cable, computer & peripheral configuration

BUS TOPOLOGY

Single Backbone Cable

ADVANTAGES

- ✓ Easy to add nodes
- ✓ Less cable than star
- ✓ Cheaper; good for small networks

DISADVANTAGES

- ✗ Main cable break = whole network down
- ✗ Needs terminators at both ends
- ✗ Low security; not for heavy traffic

STAR TOPOLOGY · MOST COMMON

Central Hub/Switch

ADVANTAGES

- ✓ Easy install & wiring
- ✓ One node fail = others unaffected
- ✓ Easy fault detection; easy add/remove

DISADVANTAGES

- ✗ Requires large cable length
- ✗ Hub failure = entire network down

RING TOPOLOGY

Closed Loop Circle

ADVANTAGES

- ✓ No central device needed
- ✓ Equal access; easy install
- ✓ Easy fault detection

DISADVANTAGES

- ✗ Sequential = delays
- ✗ Single break = whole network disrupted

MESH TOPOLOGY

Every Node Connected

ADVANTAGES

- ✓ Simultaneous multi-device data
- ✓ Node failure = network unaffected

DISADVANTAGES

- ✗ Very expensive; high redundancy

- ❌ Setup & maintenance very difficult

OSI Model · 7 Layers

Open Systems Interconnection · Released **1984** by ISO

7	Application	User interface; file, print, email, database services; user access to network
6	Presentation	Syntax & grammar; data format, encryption and compression ; translates between formats
5	Session	Manages sessions/dialogues; inserts checkpoints ; resumes interrupted dialogues; security parameters
4	Transport	End-to-end delivery; divides into packets ; resequences at destination; flow & error control
3	Network	Routing ; translates logical → physical addresses; manages traffic, switching, congestion
2	Data Link	Assembles into frames ; converts frames↔bits; physical addressing; error/flow/access control · single link
1	Physical	Bit stream over physical medium; electrical/mechanical/timing specs; data rate, synchronization; hardware-specific

*Top-Down Mnemonic:
All People Seem To Need Data*

*Bottom-Up Mnemonic:
Please Do Not Throw Sausage*

Processing

Pizza Away

Switching & Routing

CIRCUIT SWITCHING

Dedicated Circuit

Physical path **exclusively reserved** for one connection for its entire duration. Released after transfer.

Example: **Public Telephone Network (PTCL)**

- ✓ Efficient, fewer errors, reliable
- ✗ Bandwidth wasted (e.g. when listening)
- ✗ Complex setup; slow circuit establishment

PACKET SWITCHING

Data in Packets

Data split into **packets** with headers (routing info). Packets independent, can take different paths. Path **shared** among users. **Connectionless**.

Example: **The Internet**

- ✓ Efficient resource use; handles variable rates
- ✗ Packet reordering takes time
- ✗ Not good for small data packages

SOURCE ROUTING

Source node specifies entire route before transmission. Used to troubleshoot and improve performance.

HOP-BY-HOP ROUTING

Source does NOT know full path. Each node passes packet to **next node only**. Based on channel availability.

Protocols · Analog/Digital · NIC

PROTOCOL

A **set of rules and procedures** governing communication.
Without it, connected devices cannot communicate.

Functions: Defines packet size ·
Numbering scheme · Error/flow control · Connection setup/termination · Data security ·
Routing algorithms ·
Communication logs

ANALOG VS DIGITAL

Analog: Continuous waveform
(e.g. human voice)

Digital: Discrete ON/OFF states (0 or 1)

Digital→Analog for transmission (modulation), converted back at receiver

Bandwidth = data transfer rate = bits per second (bps)

NETWORK INTERFACE CARD (NIC)

Hardware device that physically connects computer to network cable.
Installed in expansion slot. Provides port for cable.

Functions: Send data to host · Receive & translate incoming data to machine language · Prepare data from computer for network cable ·
Control data flow

Top 15 Exam Tips

1 **5 Components** · Know each definition + example. She defines them precisely.

2 **3 Principles** (Delivery, Accuracy, Timeliness) · Easy 3-mark question.

3 **Data Flow** · WiFi = half-duplex. Telephone = full duplex. Always include examples.

4 **Fibre-Optic** · Know all **6 advantages** + 3 disadvantages by heart.

5 **Square Wave derivation** · Know why $a_0=0$, $a_n=0$, $b_n=4A/n\pi$ (odd only).

6 **Even/Odd functions** · Even → Cosine Series; Odd → Sine Series.

7 **Period formula** · $\sin(nx) = 2\pi/n$. Calculate for any n instantly.

8 **F.T. worked example** · $e^{-at}u(t) \rightarrow 1/(a+i\omega)$. Know every step.

9 **Time-Frequency** · Wider pulse = narrower spectrum. Narrower = wider.

10 **Serial vs Parallel** · Serial = long distance. Parallel = short/internal.

11 **Async vs Sync** · Know frame structures + error detection methods.

12 **OSI Model** · All 7 layers, order, and functions. Both directions.

13 **Topologies** · All 4: advantages + disadvantages (Bus, Star, Ring, Mesh).

14 **Circuit vs Packet** · Dedicated path vs shared path. PTCL vs Internet.

15 **OFDM** = Orthogonal
Frequency Division
Multiplexing → **5G & LTE**.
Fourier in real life.

Master Reference Tables

Transmission Media

MEDIA	TYPE	DIRECTION	FREQUENCY	KEY USE
Twisted Pair	Guided	Both	-	LAN, Telephone
Coaxial	Guided	Both	-	Cable TV, CCTV
Fibre-Optic	Guided	Both	-	Internet Backbone
Radio Waves	Unguided	Omni	3KHz–1GHz	Broadcasting
Microwave	Unguided	Uni	1–300GHz	Long distance, Satellite
Infrared	Unguided	Short	300GHz–400THz	Remote controls

OSI Layers · One-Word Recall

